

Computer Science: Grade 1, Semester 1

Matter, Form and Function

Big Idea: Computer hardware and software are designed to enable specific technological applications.

CS.1.01 - students demonstrate an understanding of the importance of ICT in knowledge generation and communication. (Week 01 - Week 01)

Essential Questions: Imagine what happens if a disrupted Internet and telecommunications networks in the world?

Skills:

demonstrate Technological skills for managing knowledge, and apply Technology in daily life - inferred 10 questions about electronic picture information.

Concepts:

data – information – knowledge – Technological skills.

Evidence:

Students' state how ICT will be used to complete their activities, and can digitally share their work and ideas on the network.

Texts & References: Not yet identified

Capstone Connection: ICT is a powerful tool to help a dwelling meet many performance specifications and optimize efficiency.

Grand Challenge Connections: Increase opportunities for Egyptians to stay and work in Egypt, Increase efficient use of our land through improved use of arid areas, Reduce pollution fouling our air water and grounds

CS.1.02 - Understand what an operating system is, and how to use one for file management, application launching, and general management of hardware and software. (Week 02 - Week 02)

Essential Questions: Do you see that the operating system is an essential component of computer technology? Why?

Skills:

create a folder – copy, move, delete and search for files and folders – save and compress files - find and remove viruses.

Concepts:

operating system – application – programming language - file – folder –virus.

Evidence:

Exit ticket – preparing an activity for managing hardware and software.

Texts & References: Not yet identified

Capstone Connection: Apply knowledge when documenting Capstone process in the online Capstone Portfolio

Grand Challenge Connections: Improve Sources of Clean Water

CS.1.03 - students understand the role of word-processing applications to create an advanced, rich document. (Week 03 - Week 03)

Essential Questions: what is an electronic document?

Skills:

create, save, format, modify and print a document.

Concepts:

text document – paragraph - word processing application.

Evidence:

electronic formatted Exit ticket.

Texts & References: Not yet identified

Capstone Connection: Word processing is an important component of curating a capstone portfolio

Grand Challenge Connections: Address the exponential population growth and prepare for the impact, Increase opportunities for Egyptians to stay and work in Egypt

CS.1.04 - students use power point program for handling interrelated information and showing in an interesting presentation. (Week 04 - Week 04)

Essential Questions: Which is the most important from your point of view to have a good presentation (show impressive - care about the presenter- goal of the show)

Skills:

create a blank & template presentation, identify slide types, control slide view, save presentation with different formats and display a presentation.

Concepts:

slide – presentation

Evidence:

preparing a presentation illustrates effect of technological advancement in getting information.

Texts & References: Not yet identified

Capstone Connection: Curation is an important part of communicating your capstone results.

Grand Challenge Connections: Address the exponential population growth and prepare for the impact, Improve the use of alternative energies to reduce our reliance on extracted fuel sources, Reduce urban congestion and its impact, Increase opportunities for Egyptians to stay and work in Egypt, Work to eradicate public health issues/disease, Improve Sources of Clean Water

CS.1.05 - students understand how to use spread sheet in the preparation of statistical tables and enhance data processing using functions and formulas to fit more than one background knowledge to the user. (Week 05 - Week 05)

Essential Questions: how could you collect, organize and analyze data to identify solutions for problems?

Skills:

input different types of data, represent data graphically, and apply some functions / formulas to analyze data.

Concepts:

spread sheet – worksheet – workbook – charts – function – formula.

Evidence:

preparing a worksheet to process data of class election and report a result.

Texts & References: Not yet identified

Capstone Connection: Apply knowledge during Prototype test design, implementation and analysis; use to capture and document information in the Capstone Portfolio; apply during the Capstone Exhibition process

Grand Challenge Connections: Increase opportunities for Egyptians to stay and work in Egypt, Increase industrial base for Egypt

Applications: ST.3.02,

CS.1.06 - students demonstrate understanding of how to select and edit digital images that illustrate theme or event. (Week 06 - Week 06)

Essential Questions: what are your experiences when dealing with pictures?

Skills:

search for picture files, use application to format, save pictures, convert 2d pictures to 3d and provide alternative applications for processing pictures.

Concepts:

Multimedia

Evidence:

search for pictures with different extensions and compare between them (size – resolution – processing programs)

Texts & References: Not yet identified

Capstone Connection: Apply knowledge and skills to Capstone process: during Prototype design, testing and analysis; documenting in Capstone Portfolio and include in Capstone Exhibition

Grand Challenge Connections: Improve the use of alternative energies to reduce our reliance on extracted fuel sources, Reduce pollution fouling our air water and grounds

Applications: DE.2.05, BI.3.06, CH.3.05,